

Computer Networks Laboratory

(CSDC-0236)

Laboratory Assignment File

B.Tech (CSE) 4th Semester

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Lab 01

Assignment (a) Introduction to Cisco Packet Tracer and Implementation of a Small Network.

Description: -

Computer networking plays a vital role in modern communication systems by enabling data exchange, resource sharing, and internet connectivity. To understand networking concepts effectively, learners must be familiar with both theoretical tools and physical infrastructure. Network simulation software helps visualize and analyze data flow, while physical cabling ensures real-world connectivity. Cisco Packet Tracer is a network simulation and visualization software designed to help students and beginners understand computer networking concepts without requiring physical networking hardware. The primary purpose of this tool is to create a virtual networking environment where routers, switches, computers, and servers can be configured and connected logically. By simulating real network behavior, it allows learners to understand how data packets move from a source device to a destination device through various networking devices and protocols.

Downloading and Installation of Cisco Packet Tracer:

1. Open any web browser and go to the official Cisco Networking Academy website.
2. If you do not have an account, create a free account by registering with your email ID and basic personal details. If you already have an account, log in using your credentials.
3. After logging in, navigate to the Resources or Downloads section available on the dashboard.
4. Locate Cisco Packet Tracer in the list of available software and learning tools.
5. Select the appropriate version of Cisco Packet Tracer based on your operating system, such as Windows (64-bit), Linux, or macOS.

6. Click on the Download button to start downloading the installer file.
7. Ensure a stable internet connection during the download process to avoid file corruption.
8. Once the download is complete, the installer file will be saved in the system's default Downloads folder.
9. Verify that the downloaded file has the correct file extension (for example, .exe for Windows).
10. The Cisco Packet Tracer installer is now ready to be installed on the system

Analytical Working of Cisco Packet Tracer:

Cisco Packet Tracer operates by simulating network behavior at both the logical and physical levels. Users design a network topology by placing devices and connecting them using appropriate cables. Each device can then be configured using command-line or graphical interfaces similar to real networking equipment. The simulation mode allows users to observe packet transmission step by step, making it possible to analyze delays, routing paths, and protocol behavior. This analytical capability helps identify configuration errors and understand the cause-and-effect relationship between network settings and performance. From an educational standpoint, Cisco Packet Tracer enhances conceptual understanding by transforming abstract networking theories into visible processes. It enables students to test different network scenarios, troubleshoot errors, and improve logical thinking. Practically, it helps in planning network designs before real deployment, thereby reducing hardware costs and configuration mistakes. Although it cannot fully replace real devices, it serves as an effective bridge between theory and hands-on practice.

Assignment (b): Hands-on with Network Physical Components for Designing a LAN Cable and Enabling Data Transmission by Connecting Two Computers..

Description: -

RJ-45 Connector and Ethernet Cabling Overview:

RJ-45 connectors are standard connectors used to terminate Ethernet cables for wired network communication. These connectors are commonly used with twisted-pair cables such as Cat5, Cat5e, and Cat6 in local area networks. Each Ethernet cable contains eight color-coded wires arranged in four twisted pairs, which help minimize electromagnetic interference and ensure reliable data transmission. The RJ-45 connector acts as the interface between network devices and the physical transmission medium.

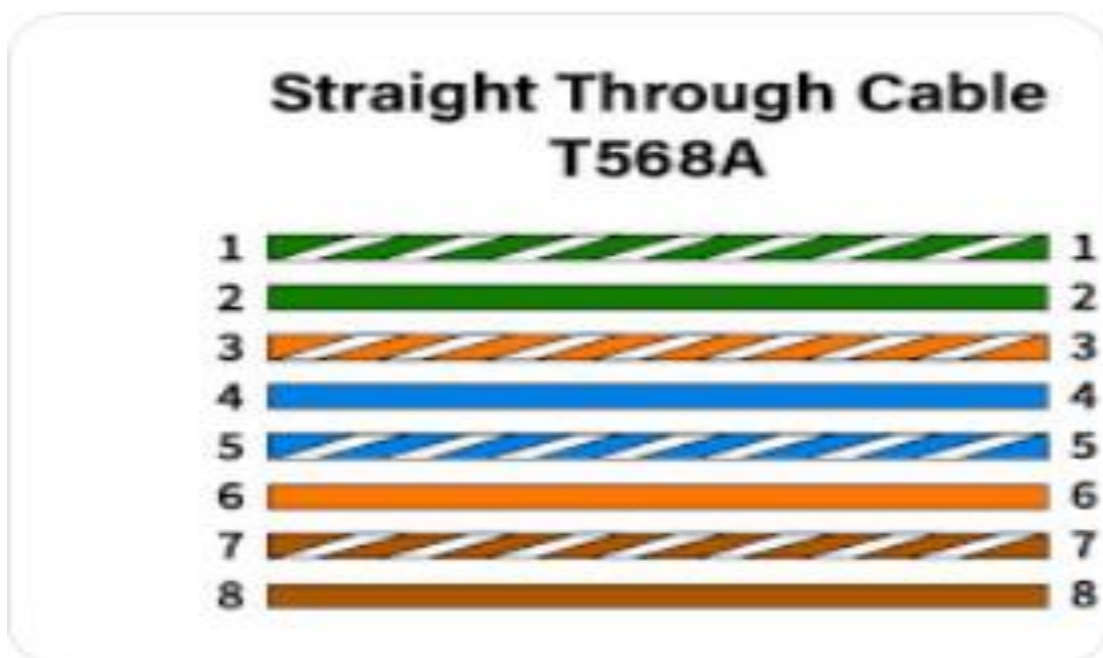


Figure1.1 : Color code Diagram for connecting the wires properly.

The preparation of an RJ-45 connector wire is a precise physical process that directly affects network performance. The outer insulation of the Ethernet cable is carefully

removed to expose the internal wires without damaging them. These wires are then straightened and arranged according to a specific wiring standard, most commonly T568B. Once aligned, the wires are inserted into the RJ-45 connector and crimped firmly using a crimping tool. Proper crimping ensures strong electrical contact between the cable wires and the connector pins.

Analytical Importance of RJ-45 Cable Preparation:

The quality of RJ-45 cable preparation determines the reliability of network communication. Incorrect wire sequencing can prevent devices from communicating, while loose crimping may cause intermittent connectivity and packet loss. These issues demonstrate the importance of the physical layer in the OSI model, as errors at this level affect all higher layers. An analytically prepared RJ-45 cable ensures stable signal transmission, reduced data loss, and improved overall network efficiency.

Relationship Between Simulation and Physical Networking

Cisco Packet Tracer and RJ-45 connector preparation complement each other in networking education. While Packet Tracer focuses on logical configuration and protocol analysis, RJ-45 cable preparation emphasizes physical connectivity and signal integrity. Together, they provide a holistic understanding of networking by linking virtual simulations with real-world hardware implementation. This integrated learning approach strengthens both analytical thinking and practical skills.

Lab 02

Assignment: Design and Implement Bus, Ring, Star, Mesh and Hybrid Network Topologies Using Cisco Packet Tracer, and Analyze their Connectivity.

Tool: CISCO Packet Tracer.

Description:

Network topology refers to the physical or logical arrangement of nodes (computers, switches, routers) and communication links in a network. It determines how data flows, how devices communicate, fault tolerance, scalability, and overall network performance. Choosing an appropriate topology is crucial because it directly affects cost, reliability, maintenance complexity, and expansion capability of the network.

Here, We will discuss different types of Network Topology

Point-to-Point Topology

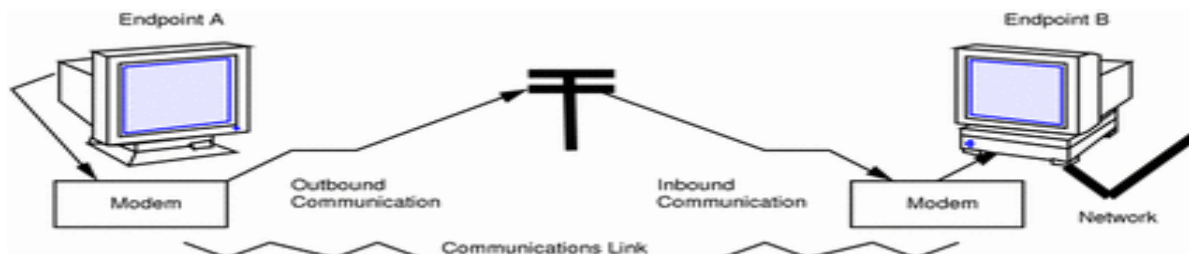
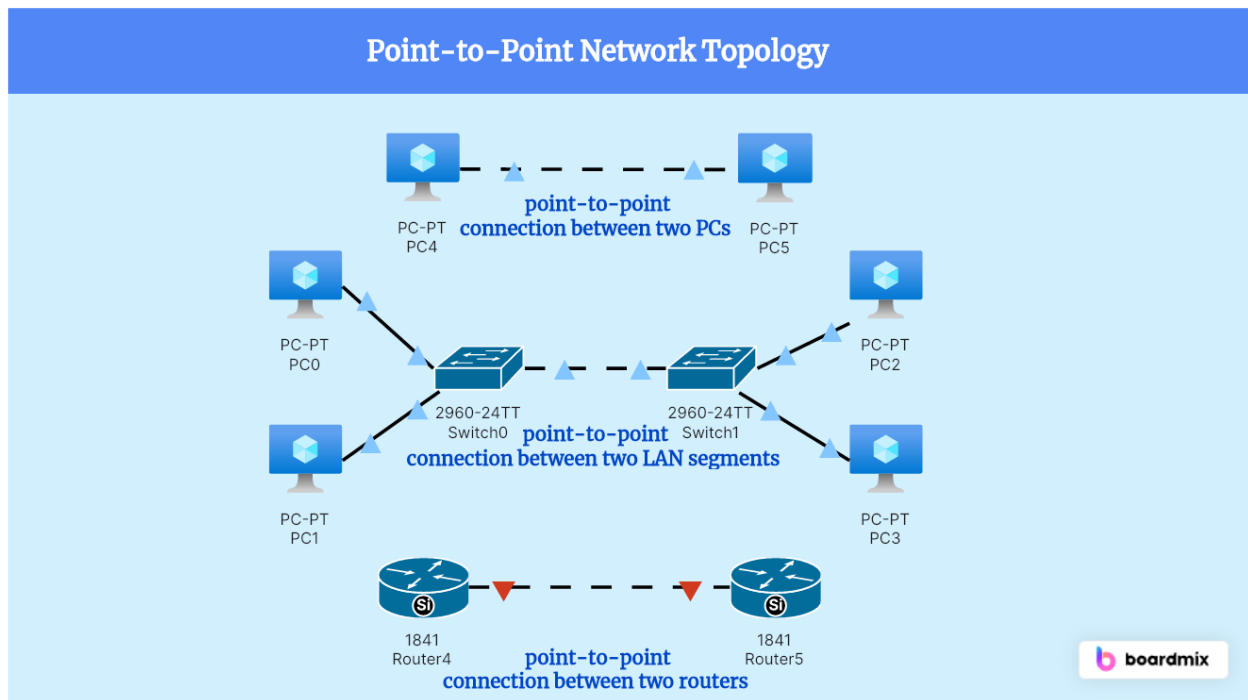


Figure 2.1: Point to Point Connection Diagram

Point-to-point topology is the simplest form of network topology, where two devices are directly connected by a single communication link. There are no intermediate devices involved in data transmission, which means data travels directly from sender to receiver without sharing the medium with other nodes. This topology is widely used in WAN links, such as leased lines, serial communication, and microwave links between two locations. From a performance perspective, point-to-point topology offers high speed, low latency, and zero collision, since the entire bandwidth is dedicated to only two devices. It is also highly secure, as data is not broadcast to

multiple nodes. However, this topology lacks scalability; adding more devices requires separate physical links, which significantly increases cost and infrastructure complexity. Additionally, if the single link fails, communication completely breaks down, making fault tolerance very low.

Star Topology:

In star topology, all devices are connected to a central device, such as a hub or switch. Each node communicates with other nodes through the central device, not directly. This structure simplifies network management and troubleshooting, as failure of a single node does not affect the rest of the network.

Analytically, star topology provides high reliability and easy scalability, since new devices can be added without disrupting existing connections. Performance is generally good, especially when a switch is used instead of a hub, because collisions are minimized. However, the central device becomes a single point of failure. If the hub or switch fails, the entire network becomes non-functional. Cost is also higher compared to bus topology due to increased cabling and hardware requirements.

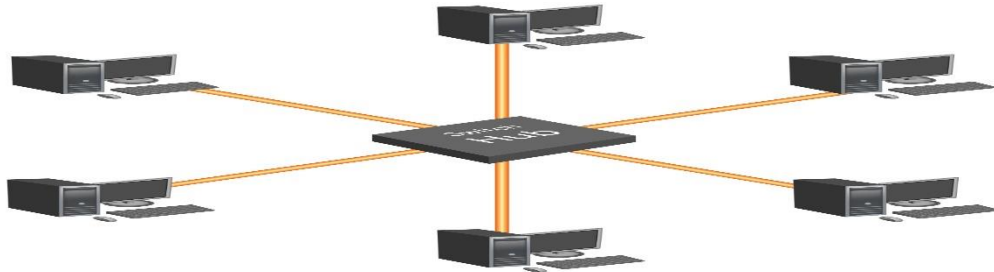


Figure 2.2: Star Topology Diagram

Bus Topology:

Bus topology uses a single shared communication medium (backbone cable) to which all devices are connected. Data sent by one device travels along the bus and is received by all nodes, but only the intended recipient processes it. This topology is cost-effective and easy to implement, making it suitable for small or temporary networks. However, as the number of devices increases, network performance degrades significantly due to collisions and congestion. Fault isolation is difficult,

and failure of the backbone cable causes complete network shutdown. Because of poor scalability and reliability, bus topology is rarely used in modern networks.

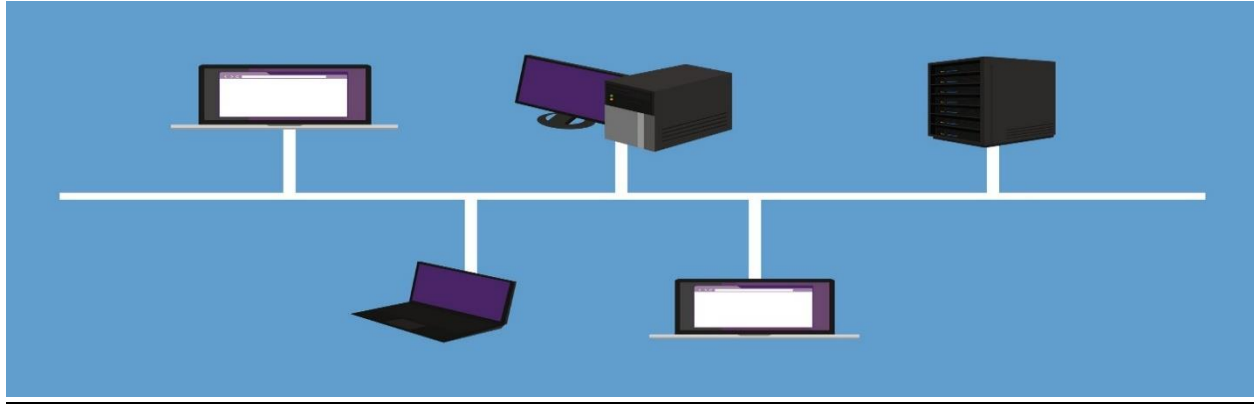


Figure 2.3: Tree Topology Diagram

Ring Topology:

In ring topology, each device is connected to exactly two neighboring devices, forming a closed loop. Data travels in one direction (or both directions in dual-ring setups) using techniques like token passing, which prevents data collisions.

Ring topology provides predictable performance, especially under heavy load, since each device gets equal access to the network. However, failure of a single node or link can disrupt the entire network, unless fault-tolerant mechanisms are implemented. Adding or removing devices is complex and may require network downtime. Due to these limitations, ring topology is largely obsolete today.

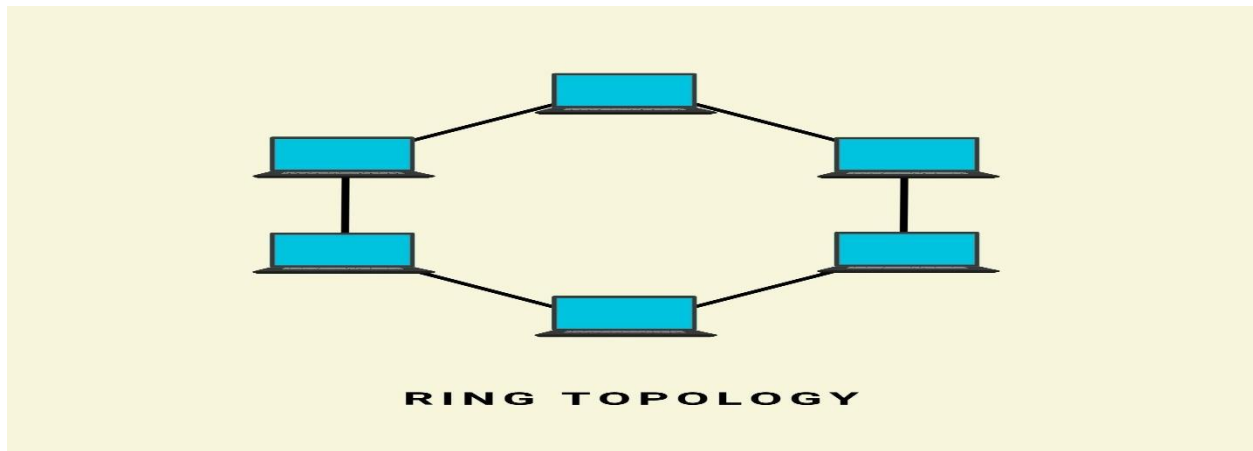


Figure 2.4: Ring Topology Diagram

Mesh Topology:

Mesh topology connects each device to multiple or all other devices in the network. In a full mesh, every node is directly connected to every other node, whereas in a partial mesh, only critical nodes have multiple connections. This topology offers exceptional reliability, fault tolerance, and redundancy, as multiple paths exist for data transmission. If one link fails, data can be rerouted through another path. Mesh topology is commonly used in backbone networks, military communication, and wireless mesh systems. However, the major drawback is high cost and complexity, as the number of links increases exponentially with the number of nodes, making it impractical for large networks.

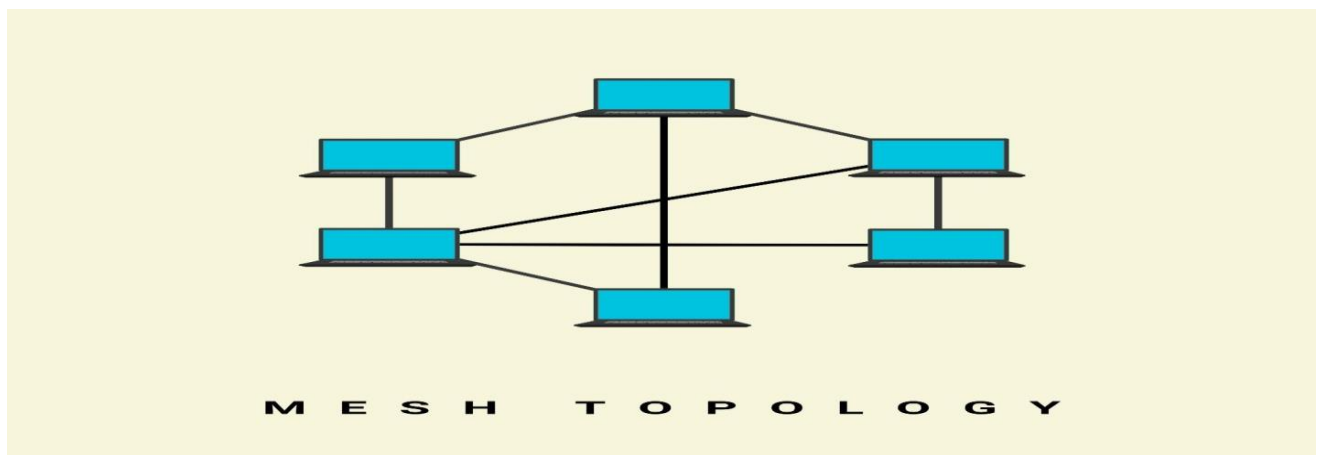


Figure 2.5: Ring Topology Diagram

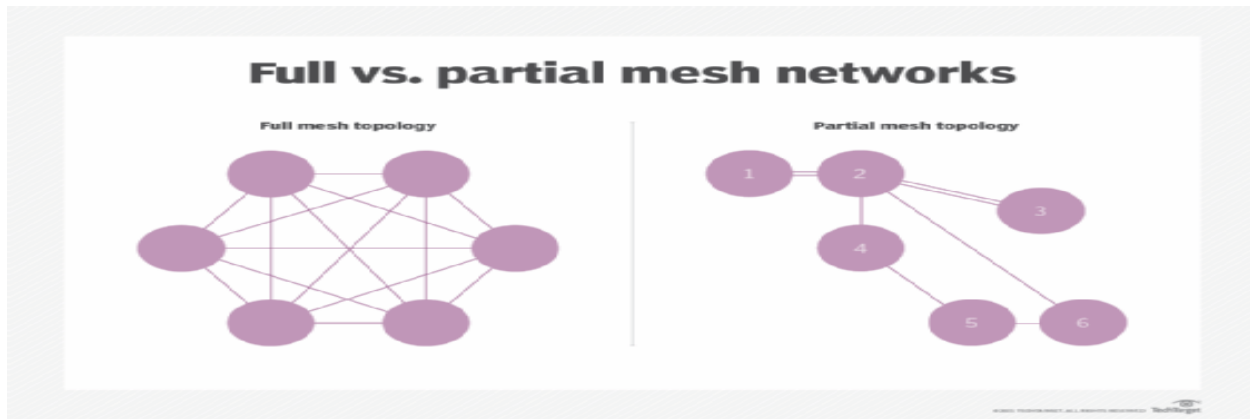


Figure 2.6: Full vs partial Mesh Topology Diagram

Tree (Hierarchical) Topology:

Tree topology is a hybrid topology that combines characteristics of star and bus topologies. Devices are arranged in a hierarchical structure, with a root node connected to intermediate nodes, which in turn connect to leaf nodes. This topology is highly scalable and well-structured, making it suitable for large organizations, campuses, and enterprise networks. Network management becomes easier due to clear hierarchy and segmentation. However, failure of higher-level nodes can affect large portions of the network. Installation and maintenance costs are also higher compared to simpler topologies.

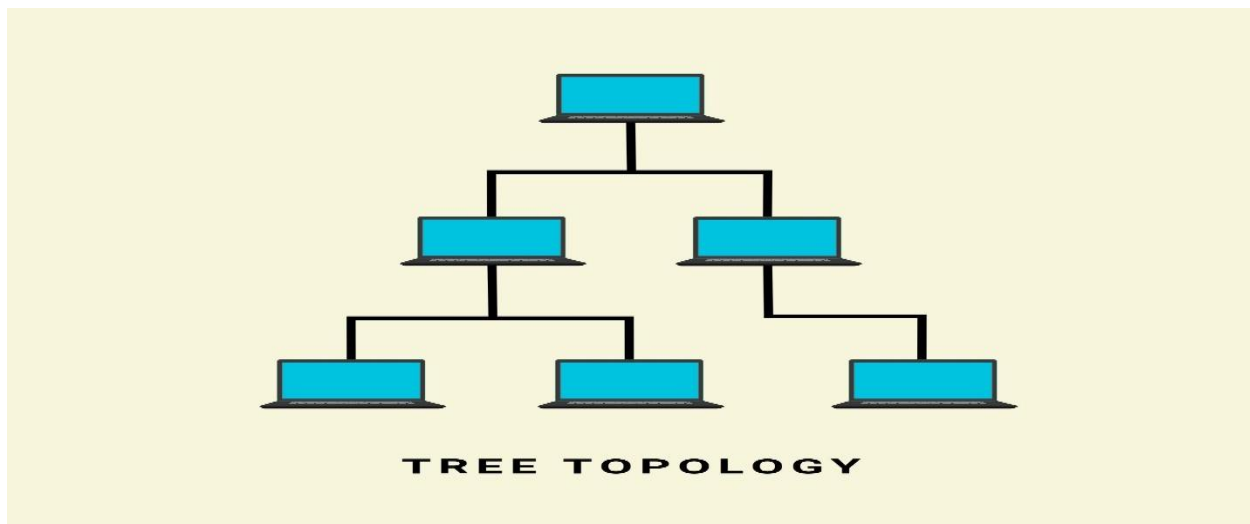


Figure 2.7: Tree Topology Diagram

Hybrid Topology:

Hybrid topology is a combination of two or more different topologies, designed to leverage the strengths of each while minimizing their weaknesses. For example, a network may use star topology within departments and bus or ring topology between departments. Analytically, hybrid topology offers high flexibility, scalability, and reliability, making it ideal for modern complex networks. However, design, implementation, and troubleshooting require advanced planning and expertise, and the cost can be significantly higher.

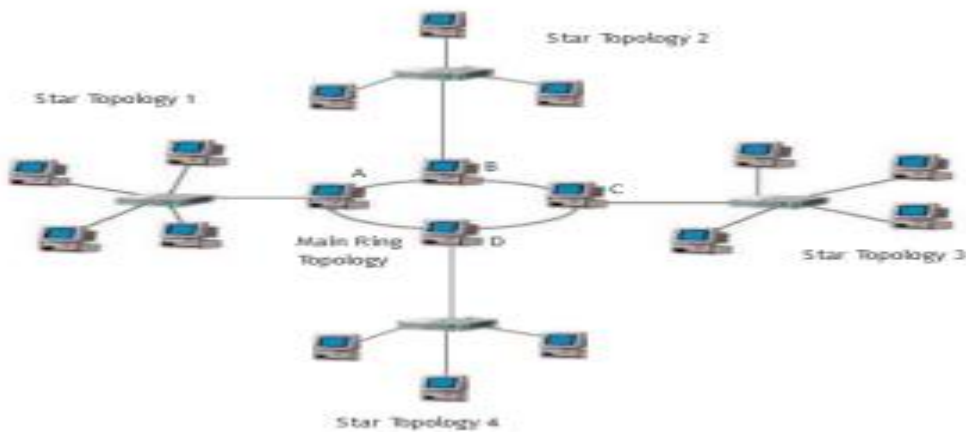


Figure 2.8: Hybrid Topology Diagram

Here, we will also attach the screenshot of our network topology using the CISCO Packet Installer simulator.

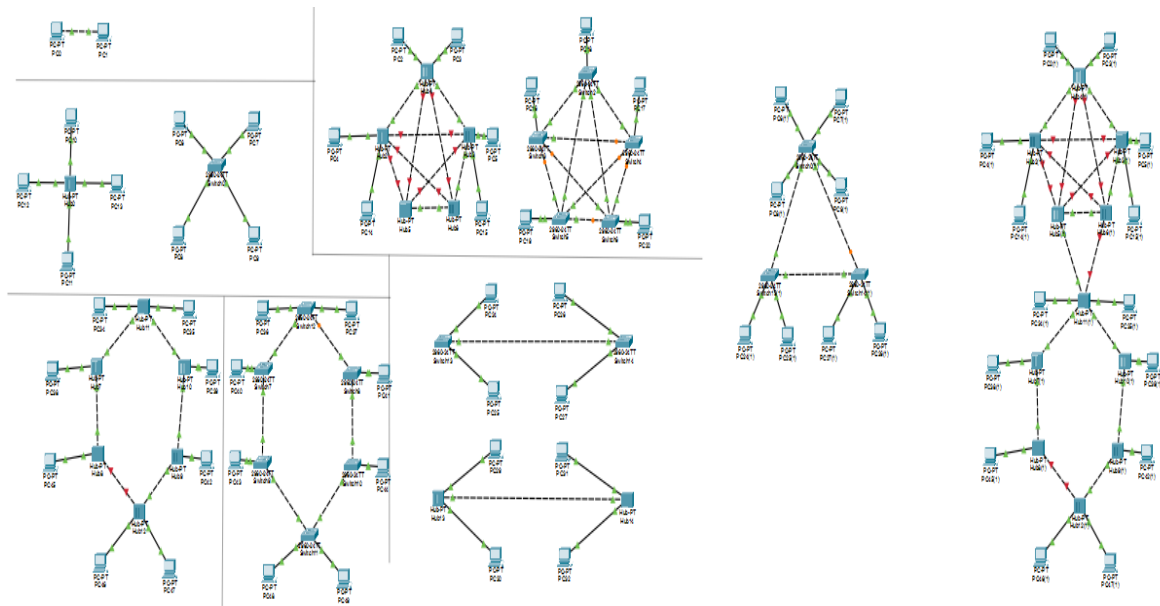


Figure2.9: Simulation Diagram of different topologies with switches and Hub.

Lab 03

Assignment: Connect two networks into a single network using a router and cables.

Tools: Cisco Packet Tracer, 1 router, switches, pc and laptop etc. virtual components.

Description:-

Network:

A computer network is an interconnected system of two or more computing devices that are linked together using communication channels in order to exchange data and share resources. These devices may include computers, laptops, servers, printers, smartphones, and other digital equipment. The primary purpose of a network is to enable communication between devices and to allow efficient sharing of information, hardware, and software resources. Networks have become a fundamental part of modern life, supporting services such as email, cloud computing, online banking, video conferencing, and access to the Internet. A computer network consists of several important components. Nodes are the devices connected to the network, such as computers and servers. Transmission media refers to the physical or wireless paths through which data travels, including Ethernet cables, optical fiber, and wireless radio signals. Networking devices such as hubs, switches, routers, and modems control data flow within and between networks. Protocols are standardized rules that govern communication; examples include TCP/IP, HTTP, FTP, and DNS. These protocols ensure that data is transmitted accurately, securely, and efficiently between devices.

Switch:

A network switch is a networking device used within a Local Area Network (LAN) to connect multiple devices and allow them to communicate with each other efficiently. Unlike a hub, which blindly broadcasts data to all connected devices, a switch intelligently forwards data only to the intended destination. This makes switches faster, more secure, and more efficient, which is why hubs are now mostly obsolete. A switch operates mainly at the Data Link Layer (Layer 2) of the OSI model, although some advanced switches operate at Layer 3. The key working principle of a switch is based on MAC (Media Access Control) addresses. Each device connected to a switch has a unique MAC address. When data arrives at a switch, it reads the source MAC address and stores it in a MAC address table along with the port number. When the destination MAC address is known, the switch forwards the data frame only to the corresponding port. If the destination MAC address is unknown, the switch temporarily broadcasts the frame to all ports to discover it. Switches significantly reduce network congestion by preventing unnecessary data transmission. Because data is sent only to the intended device, collisions are minimized, and network performance improves. Switches also support full-duplex communication, meaning data can be sent and received simultaneously, further increasing efficiency.

Routers:

A router is a networking device that connects two or more different networks and forwards data packets between them. Its primary role is to determine the best possible path for data to travel from a source network to a destination network. Routers are essential for Internet connectivity, as they connect local networks to wide area networks such as the Internet. Routers operate at the Network Layer (Layer 3) of the OSI model and make routing decisions based on IP addresses rather than MAC addresses. When a data packet arrives at a router, it examines the destination IP address and consults its routing table. The routing table contains information about known networks and the paths to reach them. Based on this information, the router forwards the packet to the next appropriate network device until it reaches its final destination. Modern routers perform many additional functions beyond routing. They provide Network Address Translation (NAT), which allows multiple devices

in a private network to share a single public IP address. Routers also act as DHCP servers, automatically assigning IP addresses to devices in a network. Most home routers include built-in firewalls to protect the network from unauthorized access and cyber threats.

Procedure to connect two different network into one single network:

1.Physical Cabling:

Connecting Devices In Cisco Packet Tracer, select the "Connections" bolt icon. Then PC to Switch: Copper Straight-Through Cable. Use Switch to Router: Copper Straight-Through Cable. Router to Router: Cross-Over (if no serial ports).

2.Desktop > IP Configuration:

Click on the PC or the laptop Interface, go to the Desktop tab, and select IP Configuration. Now manually assign all the IP Addresses like 192.168.10.01 to 06 for network 1 and 192.168.20.01 to 06 for network 2. Subnet address for both network 1 and 2 will be same as 255.0.0.0 .The Default Gateway must match the IP address we will assign to the router which is initially 0.0.0.0 and we have to assign an unique IP Address as 192.168.10.7 which is unique for both network 1 and 2.

Step 3: Accessing the Router CLI:

Go to Command Line Interface and Click the Router, navigate to the CLI tab. If asked to enter the initial configuration dialog, type no.

Step 4: Configuring Gateway 1 (G0/0):

We must assign the IP address 192.168.10.7 to the interface connecting to LAN 1.

We should see a message saying: %LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up.

Step 5: Configuring Gateway 2 (G0/1)

Now we configure the interface for the second network (192.168.20.0).

We must assign the IP address 192.168.20.7 to the interface connecting to LAN 2.

We should see a message saying: %LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up.

Step 6: Verification & Testing:

1. Check Interface Status: On the Router, check if IPs are set correctly.
2. The Ping Test: Open PC 1 (192.168.10.1) Command Prompt. Write ping 192.168.20.4

If the ping succeeds, routing is working.

Diagram:

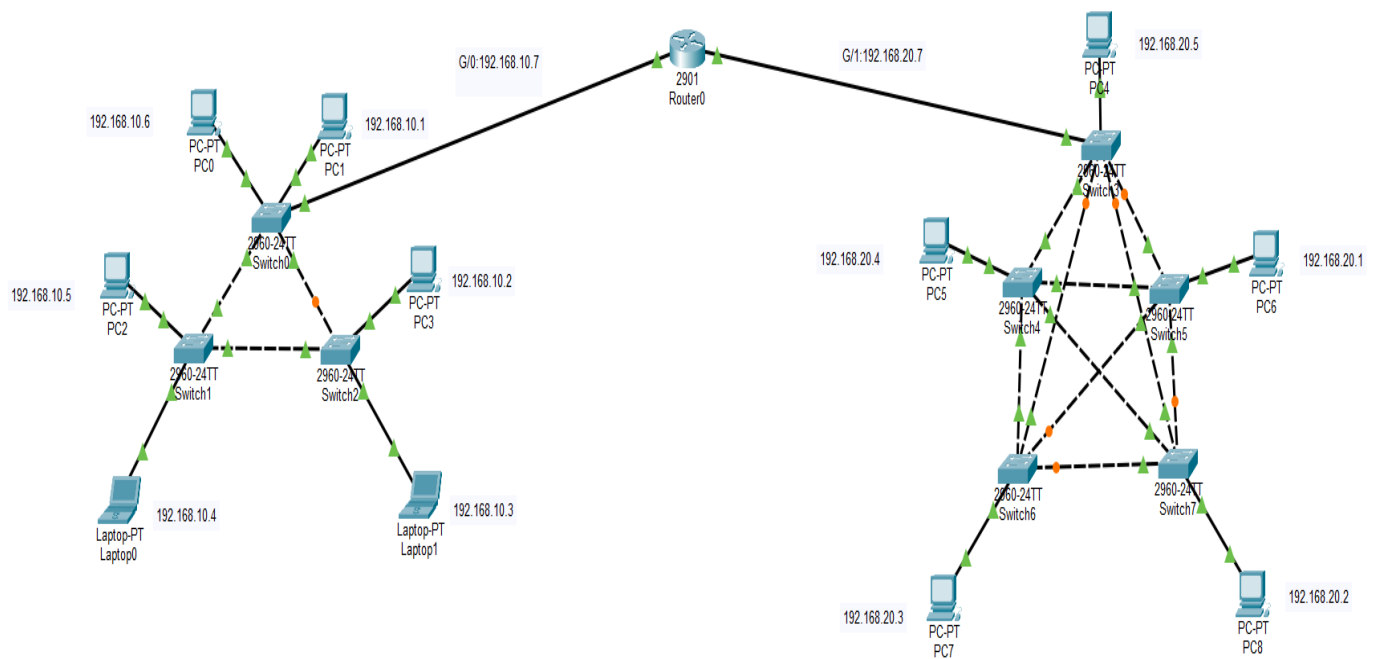


Figure3.1: Simulation Diagram of two Networks connected with a Switch.

Lab 04

Assignment: Create a scenario where two different departments (IT, HR) of different local area networks are directly connected through intermediate devices. Each group has at least 4 end systems. Each topology has at least one router, and each department is connected. Do the following for the given scenario:

- a) Provide the Static IP Address to every end device available in the network.**
- b) Label each and every interface with their IP Address.**
- c) Use a different topology for each department.**
- d) Configure static routing.**
- e) Discuss the concept of the gateway and provide the default gateway address to devices.**
- f) Show the data transmission between each department.**
- g) Also, explain the overall topology of the network.**

Tools: Cisco Packet Tracer, 1 router, switches, pc and laptop etc. virtual components.

Description:-

Our aim is to design and implement a network scenario in Cisco Packet Tracer where two different departments (IT and HR) are connected through intermediate networking devices using static IP addressing and static routing, and to demonstrate communication between them.

Network Scenario Description:

In this experiment, two separate Local Area Networks (LANs) are created:

Department 1: IT Department

Department 2: HR Department

Each department contains at least 4 end devices, One Switch and One Router. Both routers are interconnected using a serial link, allowing communication between the two departments.

Network Topology Overview:

IT Department (Star Topology):

Uses Star Topology ,One central switch connects all PCs and Router connects the LAN to external network.

Network Address: 192.168.10.0/24

HR Department (Extended Star Topology):

Uses Extended Star Topology ,Switch connects multiple PCs,Router connects HR network to IT network

Network Address: 192.168.20.0/24

Procedure of Working:

1, Open Cisco Packet Tracer and create a new file. Then start adding different required components like two 2950-24 switch and two 1841 routers and suitable cables with PC or Laptop Components.

2, Now, we need to configure each of the PC or Laptops and other components using GUI or CLI using the below format:

IT Department – 192.168.10.0/24

Device	Interface	IP Address	Subnet Mask	Default Gateway
PC0	FastEthernet	192.168.10.1	255.255.255.0	192.168.10.5

Device	Interface	IP Address	Subnet Mask	Default Gateway
PC1	FastEthernet	192.168.10.4	255.255.255.0	192.168.10.5
PC2	FastEthernet	192.168.10.2	255.255.255.0	192.168.10.5
PC3	FastEthernet	192.168.10.3	255.255.255.0	192.168.10.5
Router0	Fa0/0	192.168.10.5	255.255.255.0	—
Router0	Serial0/0/0	192.168.11.5	255.255.255.0	—

HR Department – 192.168.20.0/24

Device	Interface	IP Address	Subnet Mask	Default Gateway
PC4	FastEthernet	192.168.20.1	255.255.255.0	192.168.20.5
PC5	FastEthernet	192.168.20.2	255.255.255.0	192.168.20.5
(PC6)	FastEthernet	192.168.20.3	255.255.255.0	192.168.20.5
(PC7)	FastEthernet	192.168.20.4	255.255.255.0	192.168.20.5
Router1	Fa0/0	192.168.20.5	255.255.255.0	—
Router1	Serial0/0/0	192.168.11.4	255.255.255.0	—

3. Now we do Interface Labeling like all router interfaces are labeled as:Router0

Fa0/0 → 192.168.10.5

S0/0/0 → 192.168.11.5

Router1

Fa0/0 → 192.168.20.5

S0/0/0 → 192.168.11.4

4. Then we need to do static Routing Configuration. To enable communication between both LANs, static routing is configured.

On Router0 (IT Router):

enable

configure terminal

ip route 192.168.20.0 255.255.255.0 192.168.11.4

exit

On Router1 (HR Router)

enable

configure terminal

ip route 192.168.10.0 255.255.255.0 192.168.11.5

exit

Concept of Gateway:

A Default Gateway is the IP address of the router interface that connects a local network to other networks.

When a PC wants to communicate with a device outside its own LAN:

1. It sends the packet to the Default Gateway.
2. The router forwards it to the destination network.

Default Gateways Used:

- IT Department → 192.168.10.5
- HR Department → 192.168.20.5

Without a default gateway, devices cannot communicate outside their own LAN.

Data Transmission Between Departments

Example: PC0 (IT) → PC4 (HR)

1. PC0 (192.168.10.1) wants to send data to 192.168.20.1
2. It sends packet to its default gateway (192.168.10.5)
3. Router0 checks routing table
4. Router0 forwards packet to Router1 via serial link (192.168.11.4)
5. Router1 forwards packet to HR network
6. PC4 receives packet
7. Reply follows reverse path

Verification Command:

From PC0 write ping 192.168.20.1. The first packet sent will show failure but from the next time, it will show successful.

In this experiment, two different departmental LANs (IT and HR) were successfully designed and connected using routers and static routing. Each device was assigned a static IP address and default gateway. Communication between departments was verified using ping commands

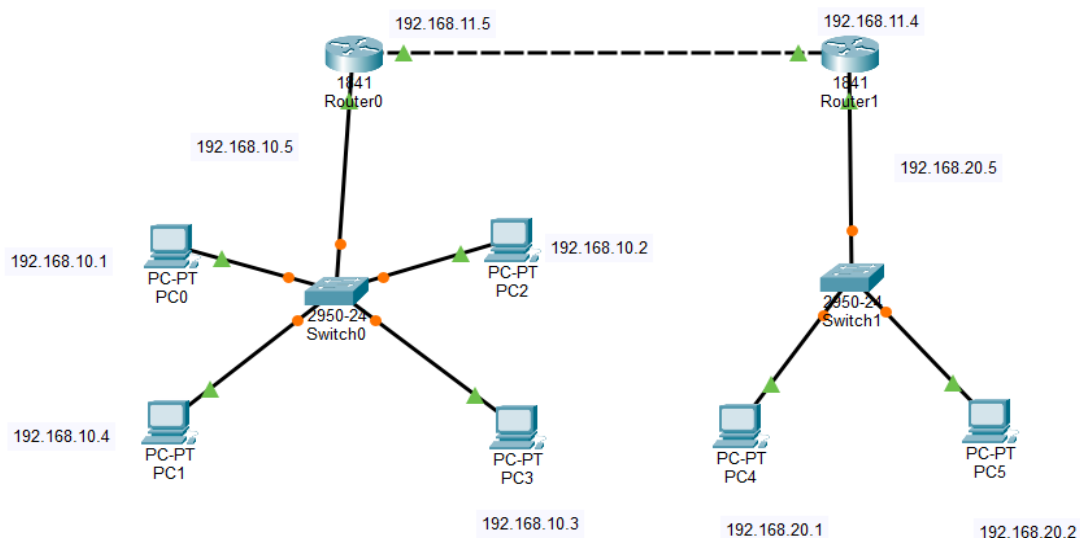


Figure4.1: Simulation Diagram for connection between two different routers

Lab -05

Assignment: Implementation of Multi-Router Network Topology using Static Routing

Objective:

The objective of this experiment is to design and implement a multi-router network using Cisco Packet Tracer. The aim is to configure IP addressing, establish connections between multiple Local Area Networks (LANs), and enable communication between them using static routing.

Tools Required: Cisco Packet Tracer

Theory:

A computer network is a system in which multiple devices are connected to share data and resources. In this experiment, we use routers, switches, and end devices.

Router:

A router is a networking device that connects different networks and forwards data packets based on IP addresses.

Switch:

A switch connects multiple devices within the same LAN and operates at Layer 2 (Data Link Layer).

IP Addressing:

Each device in a network must have a unique IP address. It is used to identify devices and ensure proper communication.

Network Description:

The network consists of 4 routers connected in a ring topology, forming redundancy and multiple communication paths.

Routers:

- Router0 (Left)
- Router1 (Top)
- Router3 (Right)
- Router2 (Bottom)

LAN Networks:

LAN 1 – 192.168.10.0/24 (Left Side)

Connected to Router0 through Switch0:

- Laptop0 → 192.168.10.1
- PC0 → 192.168.10.2
- PC1 → 192.168.10.3
- Gateway → 192.168.10.8

LAN 2 – 192.168.30.0/24 (Top Side)

Connected to Router1 through Switch2:

- PC3 → 192.168.30.1
- PC4 → 192.168.30.2
- PC2 → 192.168.30.3
- Gateway → 192.168.30.8

LAN 3 – 192.168.20.0/24 (Right Side)

Connected to Router3 through Switch1:

- PC5 → 192.168.20.1
- Laptop3 → 192.168.20.2
- Laptop4 → 192.168.20.3
- PC6 → 192.168.20.4

- Gateway → 192.168.20.8

LAN 4 – 192.168.40.0/24 (Bottom Side)

Connected to Router2 through Switch3:

- Laptop1 → 192.168.40.1
- Laptop2 → 192.168.40.2
- Gateway → 192.168.40.8

Router-to-Router Connections (Serial Links)

Connection	Network
Router0 ↔ Router1	192.175.10.0/24
Router1 ↔ Router3	192.175.20.0/24
Router3 ↔ Router2	192.175.30.0/24
Router2 ↔ Router0	192.175.40.0/24

5. IP Addressing Table:

Router0

Fa0/0 → 192.168.10.8

S0/0/0 → 192.175.10.5

S0/0/1 → 192.175.40.5

Router1

Fa0/0 → 192.168.30.8

S0/0/0 → 192.175.10.6

S0/0/1 → 192.175.20.5

Router3

Fa0/0 → 192.168.20.8

S0/0/0 → 192.175.20.6

S0/0/1 → 192.175.30.6

Router2

Fa0/0 → 192.168.40.8

S0/0/0 → 192.175.40.6

S0/0/1 → 192.175.30.5

6. Procedure:

Step 1: Create the Topology

1. Place 4 routers (1841)
2. Add 4 switches (2950-24)
3. Add PCs and laptops
4. Connect:

Router ↔ Switch → Straight cable

Router ↔ Router → Serial cable

Step 2: Configure End Devices

Example (PC0):(We need to do it for all the

- IP Address: 192.168.10.2
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.10.8

Step 3: Configure Router Interfaces

Router0 Configuration

```
enable
configure terminal
interface fa0/0
ip address 192.168.10.8 255.255.255.0
no shutdown
```

```
interface s0/0/0
ip address 192.175.10.5 255.255.255.0
no shutdown
```

```
interface s0/0/1
ip address 192.175.40.5 255.255.255.0
no shutdown
```

Router1 Configuration

```
interface fa0/0
ip address 192.168.30.8 255.255.255.0
no shutdown
```

```
interface s0/0/0
ip address 192.175.10.6 255.255.255.0
no shutdown
```

```
interface s0/0/1
ip address 192.175.20.5 255.255.255.0
clock rate 64000
no shutdown
```

Router3 Configuration

```
interface fa0/0
ip address 192.168.20.8 255.255.255.0 no shutdown
```

```
interface s0/0/0
ip address 192.175.20.6 255.255.255.0
```

no shutdown

interface s0/0/1

ip address 192.175.30.6 255.255.255.0

clock rate 64000

no shutdown

Router2 Configuration

interface fa0/0

ip address 192.168.40.8 255.255.255.0

no shutdown

interface s0/0/0

ip address 192.175.40.6 255.255.255.0

no shutdown

interface s0/0/1

ip address 192.175.30.5 255.255.255.0

clock rate 64000

no shutdown

Static Routing Configuration:

Router0

ip route 192.168.30.0 255.255.255.0 192.175.10.6

ip route 192.168.20.0 255.255.255.0 192.175.40.6

ip route 192.168.40.0 255.255.255.0 192.175.40.6

Router1

```
ip route 192.168.10.0 255.255.255.0 192.175.10.5  
ip route 192.168.20.0 255.255.255.0 192.175.20.6  
ip route 192.168.40.0 255.255.255.0 192.175.20.6
```

Router3

```
ip route 192.168.30.0 255.255.255.0 192.175.20.5  
ip route 192.168.10.0 255.255.255.0 192.175.30.5  
ip route 192.168.40.0 255.255.255.0 192.175.30.5
```

Router2

```
ip route 192.168.10.0 255.255.255.0 192.175.40.5  
ip route 192.168.30.0 255.255.255.0 192.175.30.6  
ip route 192.168.20.0 255.255.255.0 192.175.30.6
```

Verification:

Use ping command:

```
ping 192.168.20.1  
ping 192.168.40.1  
ping 192.168.30.3
```

If replies are received → Network is working correctly

Result:

All devices from different LANs were able to communicate successfully using static routing. The network topology was implemented correctly.

Conclusion:

This experiment helped in understanding:

1. Multi-router network design
2. IP addressing and subnetting
3. Static routing configuration
4. End-to-end communication

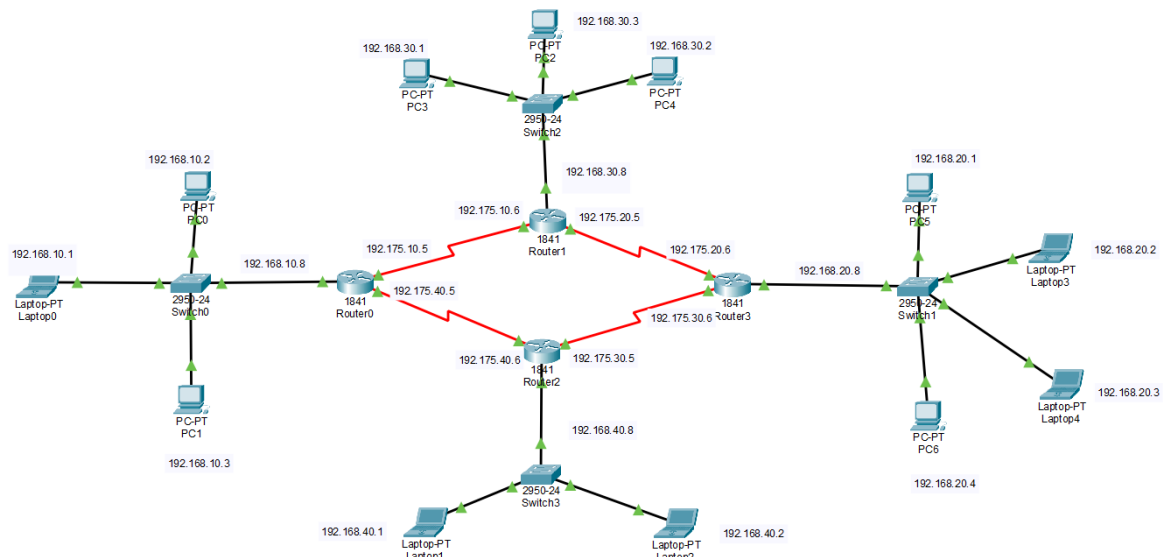


Figure5.1: Simulation Diagram for connection between system nodes and two different dynamic routers

Lab 6

Assignment: Implementation of Static Routing Between Two Networks

Tools Required: Cisco Packet Tracer

Theory:

A computer network consists of interconnected devices that communicate using IP addresses. Routers are used to connect different networks and forward data packets between them.

Static routing is a method where routes are manually configured in routers. It is simple, efficient for small networks, and provides full control over routing decisions.

Static Routing:

Static routing is a method where routes are manually configured in routers. It is simple and suitable for small networks.

Network Description:

The network consists of Two Routers

- Router4 (Left)
- Router5 (Right)

These routers are directly connected to each other.

LAN 1 (Left Side)

Network: 192.168.1.0/24

- PC0 → 192.168.1.1
- PC1 → 192.168.1.2
- Router4 (Fa0/0) → 192.168.1.5 (Default Gateway)

LAN 2 (Right Side)

Network: 192.168.2.0/24

- PC2 → 192.168.2.1
- PC3 → 192.168.2.2
- Router5 (Fa0/0) → 192.168.2.5 (Default Gateway)

Router-to-Router Linking:

192.168.1.6 and 192.168.2.8

Device IP

Router4 192.168.1.6

Router5 192.168.2.8

IP Addressing Table

Router4

Fa0/0 → 192.168.1.5

Fa0/1 → 10.0.0.1 Router5

Fa0/0 → 192.168.2.5

Fa0/1 → 10.0.0.2

Procedure:

Step 1: Create Topology

1. Place 2 routers (2811)
2. Add 2 switches
3. Add 4 PCs

4. Connect:

PC ↔ Switch → Straight cable

Switch ↔ Router → Straight cable

Router ↔ Router → Cross-over cable

Step 2: Configure End Devices

PC0:

IP Address: 192.168.1.1

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.5

PC2:

IP Address: 192.168.2.1

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.2.5

Step 3: Configure Router Interfaces

Router4 Configuration:

enable

configure terminal

interface fa0/0

ip address 192.168.1.5 255.255.255.0

no shutdown

interface fa0/1

ip address 10.0.0.1 255.255.255.252

no shutdown

Router5 Configuration:

```
enable  
configure terminal
```

```
interface fa0/0  
ip address 192.168.2.5 255.255.255.0  
no shutdown
```

```
interface fa0/1  
ip address 10.0.0.2 255.255.255.252  
no shutdown
```

Static Routing Configuration:

Router4

```
ip route 192.168.2.0 255.255.255.0 10.0.0.2
```

Router5

```
ip route 192.168.1.0 255.255.255.0 10.0.0.1
```

Verification:

Use the following commands:

```
ping 192.168.2.1
```

```
ping 192.168.2.2
```

Successful replies indicate proper routing

Conclusion:

This experiment demonstrated how static routing enables communication between two different networks. It also helped in understanding IP addressing, router configuration, and network connectivity.

Virtual Diagram:

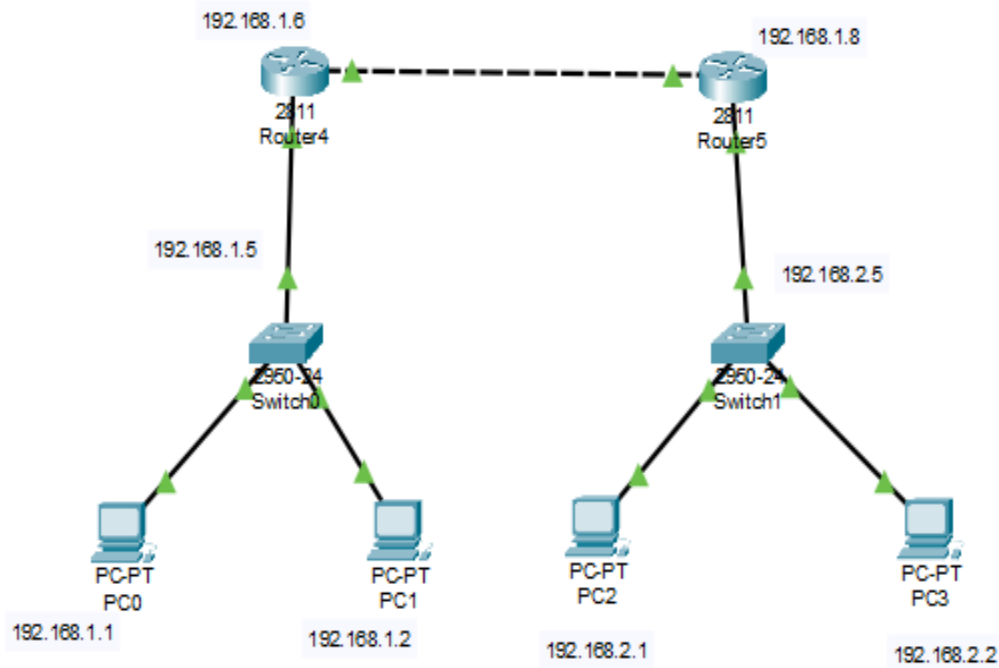


Figure6.1: Simulation Diagram for connection between system nodes and two different routers using static method.